12.471

EE25BTECH11042 - Nipun Dasari

Question:
The solution to system of equations

$$\begin{pmatrix} 2 & 5 \\ -4 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ -30 \end{pmatrix} \tag{0.1}$$

is

1) 6,2
2) −6,2
3) −6,−2
4) 6,−2

Solution:

We can write given system in the form of an augmented matrix and solve using row reduction.

$$\left(\begin{array}{cc|c} 2 & 5 & 2 \\ -4 & 3 & -30 \end{array}\right) \xleftrightarrow{R_2 \to R_2 + 2R_1} \left(\begin{array}{cc|c} 2 & 5 & 2 \\ 0 & 13 & -26 \end{array}\right) \tag{4.1}$$

$$\xleftrightarrow{R_2 \to \frac{1}{13} R_2} \left(\begin{array}{cc|c} 2 & 5 & 2 \\ 0 & 1 & -2 \end{array}\right) \tag{4.2}$$

$$\xleftrightarrow{R_1 \to R_1 - 5R_2} \left(\begin{array}{cc|c} 2 & 0 & 12 \\ 0 & 1 & -2 \end{array}\right) \tag{4.3}$$

$$\xleftrightarrow{R_1 \to \frac{1}{2} R_1} \left(\begin{array}{cc|c} 1 & 0 & 6 \\ 0 & 1 & -2 \end{array}\right) \tag{4.4}$$

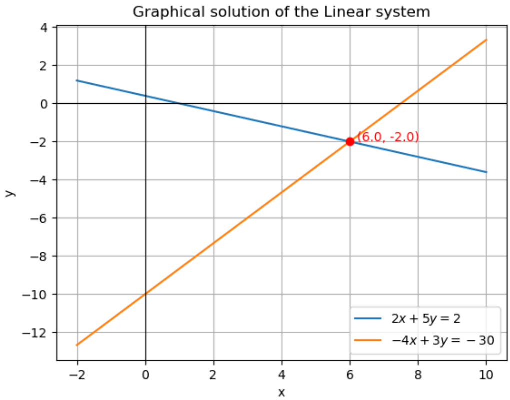
From the reduced row echelon form, we get the solution:

$$x = 6 \tag{4.5}$$

$$y = -2 \tag{4.6}$$

The solution vector is

$$\mathbf{x} = \begin{pmatrix} 6 \\ -2 \end{pmatrix} \tag{4.7}$$


Graphical solution of the Linear system
(6.0, -2.0)
$2x + 5y = 2$
$-4x + 3y = -30$
x
y
4
2
0
−2
−4
−6
−8
−10
−12
−2
0
2
4
6
8
10